

Appendix P

Detailed Analysis of Section Heading Representativeness Rating Explanations

Extremely Representative

In applying the **Extremely Representative** definition to the tool's generated descriptions, experts highlighted that the descriptions successfully incorporated every foundational theme identified within the source papers in 38 out of 60 instances. Furthermore, they noted that the descriptions provide appropriate detail beyond surface-level descriptions in 22 out of 60 cases. For instance, Expert 11 highlighted the importance of details that contextualised a topic making him establish the distinction between **Extremely Representative** and **Very Representative** regarding the section titled Electric Power System Expansion Planning:

*Because it [the description] doesn't just stop at incorporating the core topics, in this case, generation and transmission, which would be the big picture. It also offers me a bit more detail. It allows me to understand that I am going to modify the infrastructure that makes that incorporation possible. The specific detail regarding that infrastructure which is: **to meet growing and changing***

*electricity demand efficiently, reliably, and sustainably. This part is the **extra push** for me. If it [the description] had only covered transmission and infrastructure, I might have rated it as **very representative**. But now that it provides that level of detail about what that actually entails, I felt it was more representative of the role, which led me to give it a higher rating.*

In certain instances, experts compared LitTraceAI’s description to those of a human professional. In the section titled Etiology and Pathophysiology of Respiratory Sarcopenia, Expert 12 highlighted the correct inclusion of various mechanisms such as “*inflammation (chronic and acute)*” and “*mechanical ventilation,*” concluding that this was a definition expected from a specialist. Additionally, others experts emphasised the tool’s capacity to link complex ideas and provide a high level of synthesis, which is often difficult even for experts. For instance, regarding the section titled Virtual Teams in Project Management, Expert 19 noted:

This increasing interest has also generated greater attention to the benefits of virtual teams, as well as their weaknesses and the challenges associated with their management... it is valuable that the AI [LitTraceAI] has developed a coherent narrative that links the contexts of projects, knowledge management, and project success.

Although only a limited number of experts reported this observation, some evaluations within the **Extremely Representative** category highlighted LitTraceAI’s ability to create a “*coherent narrative*” that linked related concepts and contexts. Although not a pervasive finding, these instances suggest that the tool may, in certain circumstances, facilitate the synthesis of complex relationships between ideas.

The primary factor driving **Extremely Representative** ratings is the tool’s proficiency in Coverage. The most significant finding indicates the ability of LitTraceAI to capture all foundational themes and its capacity to provide precise details. These two elements combined

ensure that the output aligns closely with the expert’s expectations of a comprehensive description, consistent with the **Extremely Representative** definition provided to them. Furthermore, the LitTraceAI has shown certain potential in linking complex ideas and giving definitions expected of a specialist in the field.

Very Representative

In evaluating the output through the **Very Representative** category, experts focused primarily on the breadth of the content, noting that the description successfully captured all foundational themes in 26 of 40 instances. However, within this same thematic focus, experts noted a contrast in depth. They observed that while the core topics were present, the descriptions lacked sufficient detail in 8 cases and were identified as incomplete in 6 others.

In the case of descriptions that lacked sufficient detail, experts generally found that they remained closely aligned with the source papers, suggesting that the omission of some information did not substantially affect their overall representativeness. Incomplete descriptions, however, resulted from three distinct factors: **reliance on background sections, limitations in information availability**, and **limitations in inferential reasoning**. Collectively, these factors demonstrate how both tool design choices and inherent challenges in information interpretation influence description quality. The following paragraphs illustrate how each of these factors influenced the completeness of the generated descriptions.

First, **reliance on background sections** was identified as a limitation in four out of six cases, where the tool relied on a limited subset of paper sections. Specifically, as described in Section 4.2.1, LitTraceAI is designed to extract information from particular sections of the selected papers, such as introduction sections, literature review sections, or theoretical frameworks. Expert 15, for example, noted that LitTraceAI used only the introduction sections of the selected papers, while findings reported in the results sections could also inform the definition of participatory ergonomics, particularly where studies presented contrasting evidence:

In participatory ergonomics, studies report contrasting criteria regarding musculoskeletal disorders. For this reason, the definition should also account for these inconsistencies. The tool [LitTraceAI] often identifies concepts and statements presented in the introductions of papers. However, the results sections of some studies report evidence that contradicts those criteria.

This observation highlights a potential limitation of the tool relying on selected paper sections, as relevant or conflicting findings may also be reported elsewhere in the literature. However, this does not necessarily reflect a limitation of the tool itself, as LitTraceAI was intentionally designed to extract information only from predefined sections. Consequently, the omitted information was not available within the selected sections.

Second, some incomplete descriptions reflected information that was not available in the set of papers rather than omissions in the extracted information. This represents a **limitation in information availability**. For example, Expert 8 criticised the absence of a definition of help seeking in learning, assuming that the concept was defined in the reviewed papers. However, analysis of Brooks et al. (2015), one of the papers in the reviewed set, found that it does not explicitly define the concept in the literature review section. Instead, Brooks et al. (2015) foregrounds unresolved interpersonal dynamics associated with help seeking:

To date, little is known about the interpersonal consequences of help seeking, such as how help seeking triggers reciprocity (Mueller and Kamdar 2011). This is a surprising omission because help seeking is inherently interpersonal (Anderson and Williams 1996). In a review of the help seeking literature, Bamberger (2009) calls for a comprehensive understanding of the underlying interpersonal dynamics of help seeking.

Therefore, in this case, the incomplete description was not an error by LitTraceAI, but reflected the absence of this information in the set of papers.

Third, the findings point to **limitations in inferential reasoning** when conceptual associations are implicit rather than explicit. According to Expert 1, tactical urbanism constitutes a participatory tool due to its reconfiguration of decision-making power. This interpretation is indirectly supported by Silva (2016), one of the source papers used by LitTraceAI to generate the descriptions, which states:

In tactical actions, they [designers] are almost absent and decision-making is based on the will of non-specialised organisations and people.

While meaningful to domain experts, this inference is implicit, meaning the association between tactical urbanism and participation is not stated directly in the source text but must be inferred from the description of how decision-making power is redistributed. Because LitTraceAI relies primarily on explicit textual information, it lacks the contextual knowledge needed to make such implicit connections, illustrating how expert interpretation extends beyond surface-level textual extraction.

Beyond the scope of content coverage, only one expert raised a concern regarding the representativeness of the writing, noting a single instance in which scientific terminology appeared to be used incorrectly. Nevertheless, this issue was regarded as minor because, as Expert 27 observed, the section’s core description of Citation of Retracted Publications and Epistemic Impact remains intact even if the word “*meta-analysis*” is removed:

*I mean, the only thing I’m unsure about is that the phrase **epistemic impact on meta-analysis** kind of misses the point in a small way, right? A meta-analysis is trying to summarise scientific knowledge, so really it’s just the epistemic impact on scientific knowledge, the knowledge that’s being studied through meta-analysis, or something like that. But yeah, it pretty much gets the idea. Deleting the word meta-analysis of the description would not affect its content.*

The primary factor underlying the **Very Representative** rating is LitTraceAI’s strong performance in coverage, particularly its consistent ability to identify and incorporate the

core themes from the source papers. This performance aligns closely with the definition of **Very Representative** provided to the experts. Although the findings indicate that coverage constitutes the tool’s principal strength, a secondary pattern of insufficient detail was observed in some cases. However, this limitation is not critical, as the descriptions remain closely aligned with the source papers despite the omission of certain details.

A small number of experts noted that some descriptions appeared incomplete, which is inconsistent with the **Very Representative** definition they were given. In most instances, this issue can be attributed to **limitations in information availability**. Importantly, this does not represent a deficiency of LitTraceAI, as the available source papers did not contain the information required for a more complete description. In a small number of cases, design-related factors such as **reliance on background sections** and **limitations in inferential reasoning** may also have influenced description quality. However, further research is needed to better understand their underlying role. Finally, although an isolated instance of imprecise scientific terminology was identified, this did not affect the core content of the description. Consequently, these issues did not appear to affect the overall classification of the descriptions as **Very Representative**, as reflected in the experts’ ratings.

Moderately Representative

When evaluating outputs against this category, experts identified significant gaps in the synthesis, noting in seven out of eleven instances that the description failed to fully develop key themes or omitted important information from the source material. In most of these cases (six out of seven), the issue is due to the tool’s **reliance on background sections** such as literature reviews or introductions that did not contain the information needed to produce a complete description. The remaining instance differed from the others, involving a concern about information that appeared to be missing from the description. This exception was raised by Expert 22, who noted:

Because only Gentile’s multidimensional definition is being used, he is considered solely as an author who proposes the six dimensions of customer experience.

However, the foundational definition is being overlooked, as there are other main authors like Lemon and Verhoef, who proposed the three stages of the customer experience: pre-purchase, purchase, and post-purchase. The temporal dimension is missing.

However, an examination of the set of papers provided by Expert 22 reveals that the temporal dimension is not mentioned in any of the sources. Consequently, this omission cannot be attributed to an error of LitTraceAI. Rather, it stems from a lack of information in the source material itself, reflecting **limitations in information availability**.

The incomplete nature of these descriptions is therefore characterised by a lack of depth that substantially affects their main content. For instance, Expert 1 stated:

*This explains the application of referenced comments within polygons at a site. I would have liked the word SoftGIS to be part of the heading. It would have also been pertinent to include something related to georeferencing on-site and the use of qualitative tools. Instead, it only mentions **local knowledge**, which, yes, is obviously what is being sought: **local knowledge**. However, it does not address the georeferenced aspect of the polygons or the integration of data for planning. Furthermore, there is no mention of SoftGIS or the use of qualitative tools, which are ultimately what contribute the most and from which the most information is obtained.*

In the case identified by Expert 1, it is evident that the description omitted critical elements such as “SoftGIS” and “the use of qualitative tools” which are central to the topic under discussion. Consequently, unlike cases classified as **Very Representative**, where incomplete information does not compromise the overall description, the omissions observed here significantly undermine the completeness of the description.

Furthermore, experts highlighted that in four out of eleven cases, the description misrepresented the actual findings or relationships established within the provided papers. In

two of these four cases, the information to correct these misrepresentations was missing from the introduction or literature review section, reflecting the tool’s **reliance on background sections** as its main source of information. In only one of four cases, LitTraceAI suffered from this issue. Expert 6 emphasised it:

This topic explores the role of formal and informal institutions, such as the regulatory framework and property rights, which are part of formal institutions—nothing more. Some categories were split even though they were originally part of a broader one. In other words, if it was a human speaking, he would say: this topic explores the role of formal institutions, such as regulatory frameworks and property rights.

Examination of the source papers corroborates this concern. The following excerpt from one of the source papers illustrates the claim that the tool misinterpreted about formal and informal institutional boundaries:

... the informal economy is related to a set of firms, workers and activities that operate outside formal institutional boundaries, but within informal institutional boundaries [595] [219], formal institutional boundaries are related to the regulatory and legal framework [7]...

This example illustrates that LitTraceAI failed to accurately establish associations between formal and informal institutions. The misinterpretation is relevant because formal institutional boundaries pertain only to regulations and legal rules, confirming Expert 6’s observation that the tool did not fully capture the distinction between formal and informal institutions.

In the same context, Expert 3 perceived the tool’s description as partially inaccurate, stating that “*we don’t use LLMS for information extraction in these papers*”. However, analysis of the source papers suggests that this perception was mistaken. For instance, the

papers explicitly report that LLMs were used to support information extraction, literature reviews and hypothesis generation:

Finally, large language models have shown promise in fields such as supporting literature reviews [21] and hold potential as technologies for research hypothesis generation [93] [207].

The authors of [122] experiment with relations extracted from public knowledge graphs in the general domain and across various large language models (LLMs). In [122], the authors investigate incorrect & missing class membership relations in ontologies using large language models (LLMs).

These findings indicate that, contrary to the expert’s assumptions, LitTraceAI accurately captured the applications of LLMs. This discrepancy highlights the potential for expert perceptions to diverge from textual evidence, emphasising the importance of validating descriptions against the source material.

It is important to note that the inaccuracies in the descriptions of the four cases discussed above do not compromise their overall meaning. The experts’ explanations justifying their ratings indicate that, although the core idea of each description is preserved, it is presented with some degree of imrepresentativeness.

Beyond the breadth of content, one expert noted a lack of technical representativeness, identifying an instance in which field-specific terminology was applied inaccurately, which to some extent changed the meaning of the description.

The primary factor influencing the **Moderately Representative** rating is a lack of *Coverage*, specifically driven by the high frequency of outputs that failed to comprehensively capture or fully develop the necessary thematic depth from the source papers. Experts consistently found that the tool struggled with the completeness of the description, which represented the most prevalent weakness in the evaluation. Additionally, the findings indicate that while less frequent, issues within *Academic Language Usage* further diminished the

perceived quality through the inaccurate application of technical jargon. In most of the cases, these shortcomings were associated with the tool's **reliance on background sections**, such as introductions and literature reviews, which did not contain the information required for a more comprehensive description. Experts also identified occasional misrepresentations of findings or relationships, although these generally did not alter the overall meaning of the descriptions. Ultimately, these documented deficiencies in thematic depth and scope directly align with the **Moderately Representative** definition given to the experts, as the descriptions remained partial and failed to encompass the cohesive representation of the foundational themes.

Slightly Representative

In cases meeting the definition of **Slightly Representative**, four experts highlighted instances where the description misrepresented findings, arguments or relationships established in the provided set of papers. A primary concern was the inclusion of inaccurate information that deviated from the core themes.

For instance, Expert 24 noted that the section titled Theoretical Frameworks and Measurement in Organizational Ethics featured theories that *“are not representative.”* Furthermore, the expert highlighted the inclusion of irrelevant information, specifically noting that the *“Multidimensional Ethics Scale (MES)”* is *“not a scale for measuring Unethical Pro-organisational Behaviour.”* These inaccuracies fundamentally undermine the thematic focus of the description. Unlike the **Very Representative** category, where imrepresentativeness were minor and did not obscure the overall description, the errors in this category have a more significant negative impact. These flaws introduce irrelevant data and fail to capture the essential themes of the source material, resulting in a description that does not fully reflect its intended focus.

To clarify why the descriptions were classified as incomplete, an examination of the source papers was conducted. This analysis revealed that, in all four instances, the requisite information was absent from the literature review or introduction sections. Due to the tool's

reliance on background sections, it did not have the complete information required to generate a correct description. Furthermore, providing an accurate description in these cases requires a comprehensive understanding of the subject matter, which in certain occasions only an expert is uniquely equipped to synthesise. For instance, Expert 22 mentioned :

The AI [LitTraceAI] assumed we are talking about technology acceptance here. I get the impression it [LitTraceAI] is mixing up two different topics, because the online user experience is very different from technology acceptance.

Drawing on her deep expertise in customer experience and sustainability, Expert 22 can easily identify the error. In contrast, the tool lacks a comprehensive understanding of the subject matter due to its limited information access. This results in **limitations in inferential reasoning**, which constrain its ability to infer implicit conceptual relationships and disciplinary distinctions. This case highlights the importance of how knowledge is applied: while the source papers only briefly mentioned the construct of technology acceptance, the tool incorrectly associated it specifically with chatbots.

In general, evidence indicates a strong alignment between the provided definition and the experts' justifications for this category. This is because the descriptions remain incomplete; they overlook foundational themes and contain inaccuracies that fail to capture the full scope of the set of papers.

Additional Factors Influencing Expert Evaluation

Beyond the factors directly associated with representativeness ratings, the analysis identified several additional considerations that influenced how experts interpreted and evaluated LitTraceAI's outputs. These considerations relate not only to the technology itself but also to the characteristics of the paper set, disciplinary expectations, and the practical usefulness of the generated descriptions.

Efficiency Gains. In certain cases, experts' evaluations revealed that the quality of the description could be inextricably linked to the speed of the technology. Expert 3 discusses the efficiency advantage of LitTraceAI:

I don't think a person can't find better topics. I'm saying this because with these descriptions are very detailed. So I think that this is the advantage of using LLMs [LitTraceAI] that you can pass a lot of information and come up with good summary of these topics, a person will spend maybe I don't know, days on doing this.

Through this verbatim account, it is clear that the “*advantage*” identified by the expert lies not only in the accuracy and detail of the heading descriptions but also in the efficiency gains enabled by the technology. By rapidly processing and synthesising large volumes of information, LitTraceAI can produce detailed heading descriptions in a fraction of the time required for manual analysis, thereby reducing researcher workload and accelerating the review process. In this context, efficiency gains refer to the technology’s ability to achieve comparable or enhanced analytical outputs while requiring substantially less time and effort from researchers.

Scope Limitations of the Paper Set. Experts also recognised that some perceived shortcomings reflected limitations in the paper set rather than deficiencies in LitTraceAI itself. In a few instances, related concerns were raised regarding the number of papers included. Expert 3 noted that the description “*is only based on one paper,*” while recognising that this limitation reflected “*the set of papers I gave you.*” Although the expert considered the description accurate based on the papers provided, they acknowledged that a broader set of papers would be needed to capture the topic more comprehensively. Expert 19 similarly attributed gaps to her own actions, stating that she “*didn't give you enough information*” on Critical Success Factors in Project Management, framing the issue as an unintentional oversight. She recognised that the description was correct in relation to the supplied papers but noted that additional papers should have been included, as Critical Success Factors in Project Management encompass further dimensions that were not represented in the original set.

The issues mentioned by Expert 3 and Expert 19 reflect scope limitations of the paper set, where descriptions may be accurate for the papers provided but lack completeness when the set of papers does not adequately represent the breadth and dimensions of the topic.

Paper Selection Error. Beyond the scope limitations of the paper set, an expert also pointed to a paper selection error. Expert 9 explained that the reference to COVID-19 resulted from the inadvertent inclusion of a paper that did not align with the core topic, which in turn led to thematic distortion. He emphasised that removing this paper and rerunning the process would likely eliminate the irrelevant term. This case reflects the issue of paper selection error, whereby the inclusion of out-of-scope literature can introduce themes that are not representative of the intended corpus and thereby influence LitTraceAI’s generated description.

Broader Disciplinary Expectations. A few experts evaluated the descriptions against their wider disciplinary knowledge rather than solely against the set of papers. For instance, Expert 2 agreed that the tool’s description of geometric morphometrics in crabs was appropriate given the set of papers. However, he noted that the method is widely used beyond studies of crabs and has been applied to mammals, insects, plants, and fossils. It is also used to identify and distinguish species, examine shape changes during growth, compare populations, and investigate evolutionary patterns. Therefore, a more comprehensive description would be required if the assessment were based on this wider body of literature rather than only the selected crab studies. This observation reflects broader disciplinary expectations, where a response may be accurate with respect to the assessed literature but not encompass the full range of applications recognised across the field.

Pragmatic Utility. Experts did not evaluate errors in isolation but also considered whether those errors affected the practical usefulness of the generated descriptions. The findings of the evaluation identified several aspects that warrant closer attention, particularly subtle differences in the justification across categories. Depending on the context of the analysis, these differences may exert either a minor or a substantial influence on the

assigned rating. Such cases typically involve a lack of detail, incomplete descriptions, or the inappropriate use of academic language. When experts assign these justifications, they are effectively assessing the severity of the error.

The assessment of severity involves determining whether an error constitutes a minor issue that is largely inconsequential or a major flaw that significantly undermines the evaluation. For example, in the Very Representative category, experts noted occasional reliance on background sections and limitations in inferential reasoning. However, these issues were not considered to affect the overall quality of the description and were therefore judged as minor rather than substantive flaws. This exemplifies pragmatic utility, which refers to the extent to which a description remains useful for its intended purpose despite the presence of an error.

Another example of pragmatic utility was identified in Section 6.3.2, where pragmatic utility contributed to the classification of both the *partial LitTraceAI pipeline* and *ChatGPT* as Good. Similar to the Very Representative category, these outputs were rated positively despite their shortcomings because they remained useful for their intended purpose. Although they did not fully meet scholarly standards, they provided a usable starting point that reduced initial writing effort.

Selective Acceptance. Finally, in a few cases, experts acknowledged the validity of LitTraceAI-generated headings while questioning their relevance to the primary focus of the review. Expert 4 provided an example of selective acceptance, acknowledging that certain headings identified by LitTraceAI were defensible based on the reviewed papers, even though he would not personally foreground them when writing a state of the art. This pattern was also observed elsewhere in the study. For example, Section 6.1.2 describes how Expert 15 recognised the value of a theme identified by LitTraceAI. However, he did not consider it sufficiently relevant to the main focus of the review to include it as a major theme. This illustrates how expert judgement ultimately mediates whether a tool-generated heading is adopted, adapted, or excluded.

Although some of these factors were identified in only a small number of cases, they remain relevant because they provide important context for understanding how experts interpreted and justified their evaluations. As demonstrated in the representativeness analysis, experts did not rely solely on the category definitions when assigning ratings. They also considered issues such as **information availability**, **reliance on selected paper sections**, and **limitations in inferential reasoning**. Consequently, these factors further help explain how experts contextualised their ratings, distinguished between tool limitations and limitations of the set of papers, and interpreted the significance of observed issues when evaluating LitTraceAI’s outputs.